Maths 3

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- Swap Two Numbers Without Using a Third Variable
- Print first 10 even numbers, skip numbers divisible by 4
- Fibonacci Series
- Binary to Decimal
- Pascal Triangle

num = 110 0 11

$a = 2$, $b = 3$

$a = 3$, $b = 2$

Q1:

$a = 3$, $b = 2$

temp

temp = a;

$a = 3$

$b = 2$

temp = 3 → a

→ a = b

$a = 2$

$b = 2$

temp = 3

b = temp

$a = 2$

$b = 3$

temp = 3

→ $a = 3$, $b = 2$

$a = 5$

$b = 2$

$a = a + b$

$b = a - b$

$a = a - b$

$a = 5$

$b = 3$

→ a = 2

b = 3

```
for (i=2 ; i<=20 ; i+=2){
    if (i%4 == 0) continue;
    Sout (i);
}
```

$2 \times 10 = 20$

$i = 2 \times 10$

$= 20$

→ 0 1 1 2 3 5

N=10

first = 0 , second = 1

0 1 1 2 3 0 + 1 = 1

1 + 1 = 2

~~first~~ ~~second~~ first ~~Second~~ first d second

0 1 1 2 3 5 - - - - -

~~first~~ ~~second~~ ~~second~~ second first first

Sum = 0 + 1

= 1

How to Convert Binary number to decimal Number.

Eg:- Bin_num = 1 1 0 0 1 1

5 4 3 2 1 0

1×2^0

$+ 1 \times 2^1$

$+ 0 \times 2^2$

$+ 0 \times 2^3$

$+ 1 \times 2^4$

$+ 1 \times 2^5$

num = 1 0 0 1 0 1

6 5 4 3 2 1 0

Decimal = ?

$1 \times 2^0 + 0 \times 2^1$

$+ 0 \times 2^2 + 1 \times 2^3$

$+ 0 \times 2^4 + 0 \times 2^5 + 1 \times 2^6$

$= 1 + 8 + 64$

$= 73$

Q: 1 0 0 0 0 0

6 5 4 3 2 1 0

0×2^0

$2^6 = 64$

1 0 1

2 1 0

$2^0 + 2^2 = 5$

Q: 1 1 1 0 0 1

5 4 3 2 1 0

$2^0 + 2^3 + 2^4 + 2^5$ 32 +16 +8+1

$= 57$

n = 10 | 1
21 | 0

$1\times2^0 + 0\times2^1 + 1\times2^2$

$2^0 = 1$ ×2

$2^1 = 2$ ×2

$2^2 = 4$

$2^3 = 8$ ×2

num % 10

1 0 1

$1\times2^0 + 0\times2^1 + 1\times2^2$

×2 ×2

multiplier

→ 1 1 0 0 0 0 1 1
$2^7\ 2^6\ 2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0$

$1\times2^0 + 0\times2^1 + 0\times2^2 + 0\times2^3$

$+ 0\times2^4 + 0\times2^5 + 1\times2^6 + 1\times2^7$

=

Binary_num 1 0 0 0 0 1

$1 \times 2^0 + 0 \times 2^1 + 0 \times 2^2 + 0 \times 2^3 + 0 \times 2^4 + 1 \times 2^5$

$\Rightarrow 2^0 = 1$

$1 + 32 = 33$

0th position $\rightarrow 2^0 = 1$

$2^1 = 2$ $\times 2$

$2^2 = 4$

2^3

$2 \times 2 \times 2$

$1^{st} \rightarrow$ power $= 1 \rightarrow 1 \times 2 \rightarrow 2 \times 2 \rightarrow 2 \times 2 \times 2$

3rd iteration $\rightarrow 2 \times 2 \times 2$

$\rightarrow 2^3$

2nd position

1st position

$1\ 0\ 0\ 0 \rightarrow 0^{th}$ position

$2^3\ 2^2\ 2^1\ 2^0$

3rd position

$1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 \times 0 \times 2^0$

$= 8$

```
int bin = 1000
int i = 0;  int ans = 0;
while (bin > 0) {
    int digit = bin % 10;
    ans = ans + digit × 2^i
    i++;
}
```

$i = 0$

Digit = 0

Digit $\times 2^i$

i++;

data type

double $\leftarrow$ Math.Pow

long

int

long a = 5 ;

int z = 9 ;

$a = 2$